Koneru Lakshmaiah Education Foundation
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Review- 0

Domain: AI and Autonomous Driver Assistive Systems (ADAS)

1. AI and Autonomous Systems Applications

Artificial Intelligence (AI) and Advanced Driver Assistance Systems (ADAS) are transforming modern transportation by improving road safety, reducing human errors, and supporting drivers in different driving conditions. These systems combine Computer Vision, Deep Learning, Machine Learning, Sensor Fusion, and Control Systems to understand the vehicle's surroundings, detect potential risks, and provide timely assistance to the driver.

Some major applications include:

- **Blind Spot Detection and Warning:** Detects vehicles present in the driver's left or right blind spots using cameras and computer vision, and provides visual or audio warnings to prevent unsafe lane changes.
- **Forward Collision Warning (FCW):** Detects vehicles or obstacles ahead and estimates collision risk, alerting the driver when a potential collision is detected.
- **Automatic Emergency Braking (AEB):** Detects an imminent collision and automatically applies the brakes when the driver does not respond in time.
- **Adaptive Cruise Control (ACC):** Maintains a desired vehicle speed while automatically adjusting speed to maintain a safe distance from the vehicle ahead.
- **Pedestrian Detection and Warning:** Uses computer vision and deep learning to detect pedestrians near the vehicle and alerts the driver when there is a potential safety risk.
- **Vehicle Speed Estimation:** Estimates the speed of surrounding vehicles using camera-based video analysis and object tracking to support traffic monitoring and safety applications.
- **Safe Following Distance Monitoring:** Monitors the distance between the vehicle and the vehicle ahead and provides warnings when the following distance becomes unsafe.
- **Simulation-Based ADAS Development:** Enables ADAS algorithms to be developed and tested safely in virtual environments such as CARLA without requiring a physical vehicle.

2. Limitations in the Domain

Although ADAS technologies have made significant progress, several challenges remain:

- **Limited Blind Spot Visibility:** Conventional mirrors cannot completely cover the areas beside and behind the vehicle.
- **Detection Accuracy:** Vehicles may not be detected correctly due to occlusion, poor camera angles, or complex traffic.
- **Weather and Lighting:** Rain, fog, darkness, and glare can reduce camera performance.
- **False Warnings:** Incorrect detection may generate unnecessary alerts and distract the driver.
- **Real-Time Processing:** AI-based vehicle detection requires sufficient computing power for real-time operation.
- **Vehicle Movement:** Sudden lane changes and unpredictable vehicle movements are difficult to detect accurately.

3. Future Scope

The future of AI-powered ADAS is expected to include:

The future of AI-powered Blind Spot Detection systems can include:

- **Improved Vehicle Detection:** More accurate detection of vehicles in different traffic conditions.
- **Multi-Camera Integration:** Using multiple cameras to provide better coverage around the vehicle.
- **Predictive Warnings:** Predicting when a nearby vehicle may enter the blind spot and warning the driver earlier.
- **Sensor Fusion:** Combining cameras with radar and other sensors for more reliable detection.
- **Real-Time Edge AI:** Running AI models directly inside vehicles for faster response.
- **V2X Communication:** Using vehicle-to-vehicle communication to detect nearby vehicles beyond the camera's view.

II. Proposed Project Title

VisionGuard: Blind Spot Detection & Warning

Description

The proposed project aims to develop **VisionGuard: Blind Spot Detection & Warning**, an AI-based Advanced Driver Assistance System (ADAS) that uses Computer Vision and Deep Learning to monitor the areas around a vehicle that are difficult for the driver to see. The system will detect and track nearby vehicles using a camera feed and identify whether they enter the vehicle's left or right blind-spot zones.

When a vehicle enters a blind spot, the system will provide a real-time visual or audio warning to alert the driver and help prevent unsafe lane changes. The project will use vehicle detection, object tracking, blind-spot region identification, and risk assessment techniques. The system can be developed and tested using recorded driving videos or a simulated environment, reducing the need for physical hardware while allowing testing under different traffic conditions.

III. Objectives to Achieve the Project

Objective 1

Develop a Computer Vision system to detect nearby vehicles in real time using camera input and deep learning models.

Objective 2

Track detected vehicles and identify their positions relative to the vehicle's left and right blind-spot zones.

Objective 3

Design a warning system that determines when a vehicle enters a blind spot and provides appropriate visual or audio alerts to the driver.

Objective 4

Test and evaluate the system under different traffic conditions to improve detection accuracy, response time, and overall safety.

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